

Model Derivations

Companion notes to “Trade Tariffs and Exchange Rates”

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These notes derive three models: the two-country benchmark, the two-layer nested three-country model with the reversal threshold, and the canonical competing-exporters model from the preferential-trade-agreement (PTA) literature. Prose is limited to defining variables and linking equations.

1 Two-country model

1.1 Setup

Countries $i \in \{A, B\}$. Goods: tradable T_i (produced only in i , consumed in both) and non-tradable N_i . Labor is the only factor, immobile across countries, endowment L_i .

Technology (linear):

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i. \quad (1)$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price of each tradable in its own currency) give

$$w_i = A_{T_i}, \quad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (2)$$

Let e be the nominal exchange rate: units of A 's currency per unit of B 's currency. The price of T_B in A 's currency is e ; the price of T_A in B 's currency is $1/e$.

Preferences (Cobb–Douglas, common weights):

$$U_i = C_{T_A i}^{\alpha_{T_A}} C_{T_B i}^{\alpha_{T_B}} C_{N_i}^{\alpha_{N_i}}, \quad \alpha_{T_A} + \alpha_{T_B} + \alpha_{N_i} = 1, \quad (3)$$

where C_{gi} is country i 's consumption of good g .

1.2 Demands

Cobb–Douglas expenditure shares are constant. With income I_i and consumer prices $P_g^{(i)}$ (in i 's currency),

$$P_g^{(i)} C_{gi} = \alpha_g I_i \implies C_{gi} = \frac{\alpha_g I_i}{P_g^{(i)}}. \quad (4)$$

Under free trade $I_i = w_i L_i$, $P_{T_B}^{(A)} = e$, $P_{T_A}^{(B)} = 1/e$.

1.3 Equilibrium under free trade

Country A 's trade balance in A 's currency is export revenue minus import spending. From (4), B 's purchases of T_A are $C_{TAB} = \alpha_{TA} I_B / (1/e) = \alpha_{TA} w_B L_B e$, valued at producer price 1 in A 's currency, and A 's purchases of T_B are $C_{TBA} = \alpha_{TB} w_A L_A / e$, valued at price e . Therefore

$$TB_A = 1 \cdot C_{TAB} - e C_{TBA} = \alpha_{TA} w_B L_B e - \alpha_{TB} w_A L_A. \quad (5)$$

Setting $TB_A = 0$ (by Walras' law this is the only independent condition):

$$e = \frac{\alpha_{TB}}{\alpha_{TA}} \cdot \frac{w_A L_A}{w_B L_B} = \frac{\alpha_{TB}}{\alpha_{TA}} \cdot \frac{A_{TA} L_A}{A_{TB} L_B} \quad (6)$$

1.4 Tariff

Country A levies ad valorem tariff $\tau \geq 0$ on imports of T_B . The consumer price in A becomes $(1 + \tau)e$; producers in B still receive 1. Revenue is rebated lump-sum:

$$I_A = w_A L_A + T_A^{\text{rev}}, \quad T_A^{\text{rev}} = \tau e C_{TBA}. \quad (7)$$

Using $C_{TBA} = \alpha_{TB} I_A / [(1 + \tau)e]$,

$$I_A = w_A L_A + \frac{\tau}{1 + \tau} \alpha_{TB} I_A \implies I_A = \frac{w_A L_A}{1 - \frac{\alpha_{TB} \tau}{1 + \tau}}. \quad (8)$$

The trade balance (flows valued at producer prices; the tariff wedge is a domestic transfer):

$$TB_A = C_{TAB} - e C_{TBA} = \alpha_{TA} w_B L_B e - \frac{\alpha_{TB} I_A}{1 + \tau}. \quad (9)$$

Setting $TB_A = 0$ and substituting (8):

$$e(\tau) = \frac{1}{1 + \tau} \cdot \frac{1}{1 - \frac{\alpha_{TB} \tau}{1 + \tau}} \cdot \frac{\alpha_{TB}}{\alpha_{TA}} \cdot \frac{A_{TA} L_A}{A_{TB} L_B} \quad (10)$$

Result 1.1. $e(\tau)$ is strictly decreasing in τ for $\alpha_{TB} \in (0, 1)$: a tariff appreciates the tariff-imposing country's currency.

Proof. Write $e(\tau) = e(0) \cdot f(\tau)$ with $f(\tau) = [(1 + \tau) - \alpha_{TB} \tau]^{-1} = [1 + (1 - \alpha_{TB})\tau]^{-1}$. Since $1 - \alpha_{TB} > 0$, $f' < 0$. $\square$

1.5 Relative wage, terms of trade, and real exchange rate

Relative wage. Wages in own currency are $w_A = A_{T_A}$ and $w_B = A_{T_B}$, both constant. The relative wage in a common currency (A 's) is

$$\omega \equiv \frac{e w_B}{w_A} = \frac{A_{T_B}}{A_{T_A}} e \propto e. \quad (11)$$

A rise in e (depreciation of A 's currency) is a fall in A 's wage relative to B 's.

Terms of trade. A 's terms of trade are the price of its exports relative to its imports, in a common currency and at world (pre-tariff) prices:

$$\text{ToT}_A \equiv \frac{P_{T_A}}{e P_{T_B}} = \frac{1}{e}. \quad (12)$$

Combining (11) and (12): the exchange rate, the (inverse) common-currency relative wage, and the (inverse) terms of trade are the same object up to constants,

$$e \propto \frac{w_B e}{w_A} \propto \frac{1}{\text{ToT}_A}. \quad (13)$$

This holds because producer prices are unit labor costs ($P_{T_i} = w_i/A_{T_i} = 1$): one factor and constant returns make the identification exact.

Real exchange rate. The consumer price index in each country is the Cobb–Douglas index over consumer prices (constants κ_i collect the share terms):

$$P_A = \kappa_A [P_{T_A}]^{\alpha_{T_A}} [(1 + \tau)e]^{\alpha_{T_B}} [P_{N_A}]^{\alpha_N} = \kappa_A [(1 + \tau)e]^{\alpha_{T_B}} \left(\frac{A_{T_A}}{A_{N_A}}\right)^{\alpha_N}, \quad (14)$$

$$P_B = \kappa_B [1/e]^{\alpha_{T_A}} [P_{T_B}]^{\alpha_{T_B}} [P_{N_B}]^{\alpha_N} = \kappa_B e^{-\alpha_{T_A}} \left(\frac{A_{T_B}}{A_{N_B}}\right)^{\alpha_N}. \quad (15)$$

The real exchange rate is

$$q \equiv \frac{e P_B}{P_A} = \text{const} \times e^{1 - \alpha_{T_A} - \alpha_{T_B}} (1 + \tau)^{-\alpha_{T_B}} = \text{const} \times e^{\alpha_N} (1 + \tau)^{-\alpha_{T_B}}, \quad (16)$$

so in logs

$$d \log q = \alpha_N d \log e - \alpha_{T_B} \frac{d\tau}{1 + \tau}. \quad (17)$$

Equation (17) has three implications. First, q moves with e , attenuated by the non-tradable share α_N . Second, the tariff appreciates A in real terms directly, at any e , because it raises A 's consumer price index. Third, since $P_{N_i} = A_{T_i}/A_{N_i}$ is a productivity ratio, non-tradables affect the level of q but enter its response to trade shocks only through the exponent α_N .

1.6 The Marshall–Lerner condition

Let export volume depend on the price foreign buyers face, $p^* = 1/e$, and import volume on the price home buyers face, $p = (1 + \tau)e$. Write $X(p^*)$ and $M(p)$ with elasticities

$$\varepsilon_X \equiv -\frac{X'(p^*)p^*}{X} \geq 0, \quad \varepsilon_M \equiv -\frac{M'(p)p}{M} \geq 0. \quad (18)$$

The trade balance in A 's currency, at producer prices, is

$$\text{TB}_A(e, \tau) = X(1/e) - e M((1 + \tau)e). \quad (19)$$

Response to the exchange rate. Differentiate (19) with respect to e at $\tau = 0$:

$$\frac{\partial \text{TB}_A}{\partial e} = -\frac{X'(1/e)}{e^2} - M - e M'(e) = \frac{\varepsilon_X X}{e} - M + \varepsilon_M M, \quad (20)$$

using $X' = -\varepsilon_X X e$ and $e M' = -\varepsilon_M M$. Evaluated at balanced trade ($X = eM$):

$$\boxed{\frac{\partial \text{TB}_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1)} \quad (21)$$

A depreciation improves the trade balance if and only if $\varepsilon_X + \varepsilon_M > 1$. This is the Marshall–Lerner condition. The term -1 in (21) is the valuation effect: a depreciation raises the home-currency cost of each unit of imports, which worsens the balance one-for-one. The two volume responses must offset this effect.

Response to the tariff. Differentiate (19) with respect to τ at fixed e :

$$\frac{\partial \text{TB}_A}{\partial \tau} = -e M'((1 + \tau)e) e = \frac{\varepsilon_M}{1 + \tau} e M > 0 \quad \text{whenever } \varepsilon_M > 0. \quad (22)$$

A tariff improves the trade balance at a fixed exchange rate as long as import demand slopes down. (With a revenue rebate, income effects modify (22); in the Cobb–Douglas model of Section 1.4 the producer-price import value is $\alpha_{TB} I_A / (1 + \tau) \propto [1 + (1 - \alpha_{TB})\tau]^{-1}$, still decreasing in τ .)

Why Marshall–Lerner is necessary and sufficient for the conventional wisdom. Equilibrium requires $\text{TB}_A(e(\tau), \tau) = 0$. By the implicit function theorem,

$$\frac{de}{d\tau} = -\frac{\partial \text{TB}_A / \partial \tau}{\partial \text{TB}_A / \partial e}. \quad (23)$$

The numerator is positive by (22): the tariff creates a surplus at the initial exchange rate. To restore balance, the exchange rate must move in the direction that worsens the trade balance. If Marshall–Lerner holds, that direction is appreciation, $de < 0$, which is the conventional result. If Marshall–Lerner fails, depreciation worsens the balance instead, and the same tariff-induced surplus requires $de > 0$, so the conventional result reverses. The sign of $de/d\tau$ is therefore the opposite of

the sign of $\varepsilon_X + \varepsilon_M - 1$. This is why the condition is both necessary and sufficient.

The Cobb–Douglas case. With constant expenditure shares, import volume is $M = \alpha_{T_B} I/p$, so $\varepsilon_M = 1$; similarly $\varepsilon_X = 1$. Then $\varepsilon_X + \varepsilon_M = 2 > 1$ and (21) is positive. Marshall–Lerner holds automatically, which is why the closed form (10) is monotone in τ .

Multilateral remark. In the three-country model of Section 2, each bilateral relationship satisfies a Marshall–Lerner condition, and the tariff still improves A ’s overall balance at fixed exchange rates. The bilateral conditions do not determine how the required adjustment is split between e_{AB} and e_{AC} . That split depends on substitution across foreign origins, governed by ρ , relative to substitution between home and foreign goods, governed by η . This is the subject of Section 2.

2 Three-country nested model

2.1 Setup

Countries $i \in \{A, B, C\}$, each producing tradable T_i and non-tradable N_i with the same linear technology, wage equation, and normalization as (2). Exchange rates e_{ij} (i ’s currency per unit of j ’s), $e_{ii} = 1$, no-arbitrage $e_{ij} = e_{ik}e_{kj}$; the free rates are e_{AB}, e_{AC} . Tariffs $\tau_{ij} \geq 0$ (i ’s tariff on j ’s good). Consumer price of T_j in i : $p_{ij} = (1 + \tau_{ij}) e_{ij}$.

Preferences (two-layer nest):

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} \quad (\text{Cobb–Douglas outer}) \quad (24)$$

$$C_{T_i} = \left[\alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (\text{home vs. imports, elasticity } \eta) \quad (25)$$

$$M_i = \left[\sum_{j \neq i} \beta_j^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{across origins, elasticity } \rho) \quad (26)$$

with home weight α_D , origin weights β_j (symmetric case: $\beta_j = 1/2$). The single-elasticity model is the special case $\rho = \eta$.

2.2 Price indices and expenditure shares

CES price indices (in i ’s currency):

$$P_{M_i} = \left[\sum_{j \neq i} \beta_j p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1 - \alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (27)$$

Expenditure shares of total income I_i (rows sum to α_T):

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad (28)$$

$$s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_j \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (29)$$

2.3 Income and trade balances

Tariff revenue is rebated lump-sum. A fraction $\tau_{ij}/(1 + \tau_{ij})$ of spending $s_{ij}I_i$ is revenue, so

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (30)$$

The shares depend only on prices, so (30) is a closed form.

Trade balances at producer prices (spending $s_{ij}I_i$ is tariff-inclusive; divide by $1 + \tau_{ij}$ to get the producer-price value):

$$\text{TB}_i = \sum_{k \neq i} \frac{s_{ki} I_k}{1 + \tau_{ki}} - \sum_{j \neq i} \frac{s_{ij} I_i}{1 + \tau_{ij}}, \quad (31)$$

with all quantities expressed in a common currency (this does not affect the zeros). Equilibrium: $\text{TB}_B = \text{TB}_C = 0$; $\text{TB}_A = 0$ follows by Walras' law.

2.4 Linearization at the symmetric point

Set $L_i = 1$, $A_{T_i} = A_{N_i} = 1$, $\beta_j = 1/2$, and evaluate at free trade ($\tau_{ij} = 0$), where $e_{AB} = e_{AC} = 1$ and all incomes equal 1. Baseline shares:

$$s_{ii} = \alpha_T \alpha_D \equiv a, \quad s_{ij} = \frac{1}{2} \alpha_T (1 - \alpha_D) \equiv \frac{m}{2}, \quad m \equiv \alpha_T (1 - \alpha_D). \quad (32)$$

Perturb by $d\tau$ (A 's tariff on B) and let $\nu_j \equiv d \log e_{Aj}$ ($\nu_A = 0$). Log price changes: $d \log p_{ij} = \nu_j + \mathbf{1}\{(i, j) = (A, B)\} d\tau$.

Index changes, from (27):

$$d \log P_{M_i} = \sum_{j \neq i} \beta_j d \log p_{ij}, \quad d \log P_{T_i} = \alpha_D d \log p_{ii} + (1 - \alpha_D) d \log P_{M_i}. \quad (33)$$

Explicitly:

$$d \log P_{M_A} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{M_B} = \frac{1}{2}\nu_C, \quad d \log P_{M_C} = \frac{1}{2}\nu_B. \quad (34)$$

Share changes, from (28)–(29):

$$d \log s_{ii} = (1 - \eta)(d \log p_{ii} - d \log P_{T_i}), \quad (35)$$

$$d \log s_{ij} = (1 - \rho)(d \log p_{ij} - d \log P_{M_i}) + (1 - \eta)(d \log P_{M_i} - d \log P_{T_i}). \quad (36)$$

Income changes, from (30) (the revenue term is first order for country A only):

$$d \log I_A = \frac{m}{2} d\tau, \quad d \log I_B = \nu_B, \quad d \log I_C = \nu_C. \quad (37)$$

Differentiate (31) for $i = B, C$ at the baseline. Each term $sI/(1 + \tau)$ contributes $\frac{m}{2}(d \log s +$

$d \log I - d\tau$), where $d\tau = d\tau$ appears only in the A -to- B flow:

$$dTB_B = \frac{m}{2} \left[(d \log s_{AB} + \frac{m}{2} d\tau - d\tau) + (d \log s_{CB} + \nu_C) - (d \log s_{BA} + \nu_B) - (d \log s_{BC} + \nu_B) \right] = 0, \quad (38)$$

$$dTB_C = \frac{m}{2} \left[(d \log s_{AC} + \frac{m}{2} d\tau) + (d \log s_{BC} + \nu_B) - (d \log s_{CA} + \nu_C) - (d \log s_{CB} + \nu_C) \right] = 0. \quad (39)$$

Substituting (35)–(36) and collecting terms in $(\nu_B, \nu_C, d\tau)$ gives a linear system. Define

$$D \equiv 3\alpha_D(\eta - 1) + \rho + 1, \quad \rho^* \equiv 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]. \quad (40)$$

Collecting terms yields the linear system

$$-\frac{mD}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \frac{m}{4} \begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau. \quad (41)$$

Invert using $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = -\frac{1}{3D} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau = \frac{1}{3D} \begin{pmatrix} -(\rho + \rho^*) \\ \rho - \rho^* \end{pmatrix} d\tau. \quad (42)$$

Result 2.1 (Threshold). *At the symmetric point,*

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho + \rho^*}{3D} < 0, \quad \boxed{\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\rho - \rho^*}{3D}} \quad (43)$$

with $D > 0$ for $\eta \geq 1$ (local stability). A appreciates against B for all parameters; A depreciates against the untariffed C if and only if $\rho > \rho^*$.

2.5 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$ (the single-elasticity CES). Then

$$\rho^* - \sigma = 3 + 3\alpha_D(\sigma - 1) - 3\alpha_T(1 - \alpha_D) - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T). \quad (44)$$

Result 2.2 (Impossibility). *If $\alpha_D \geq 1/3$, both terms in (44) are nonnegative and the second is positive, so $\rho^* > \sigma$ for every $\sigma > 0$: the conventional appreciation against C survives at any elasticity. At $\alpha_D = 1/3$, $\alpha_T = 3/4$ (the symmetric baseline calibration $\alpha_{T_j} = \alpha_N = 1/4$): $\rho^* - \sigma = 1/2$ and $D = 2\sigma$, so*

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\sigma - \rho^*}{3D} = -\frac{1}{12\sigma} \rightarrow 0^-. \quad (45)$$

2.6 Nominal versus real exchange rate in the three-country model

The quantities of Section 1.5 carry over: e_{Aj} is proportional to j 's relative wage in A 's currency and to the inverse of A 's bilateral terms of trade with j . The real rate against C is $q_{AC} = e_{AC}P_C/P_A$ with $P_i = \kappa_i (P_{T_i}^{\text{own}})^{\alpha_T} (P_{N_i})^{1-\alpha_T}$, where $P_{T_i}^{\text{own}}$ is the tariff-inclusive tradable index (27) in i 's own currency and $P_{N_i} = A_{T_i}/A_{N_i}$ is constant. Hence

$$d \log q_{AC} = d \log e_{AC} + \alpha_T (d \log P_{T_C}^{\text{own}} - d \log P_{T_A}^{\text{own}}). \quad (46)$$

At the symmetric point, converting the indices to own currency ($d \log P_{T_C}^{\text{own}} = d \log P_{T_C} - \nu_C$):

$$d \log P_{T_A}^{\text{own}} = (1 - \alpha_D) \frac{1}{2} (\nu_B + \nu_C + d\tau), \quad (47)$$

$$d \log P_{T_C}^{\text{own}} = \alpha_D \nu_C + (1 - \alpha_D) \frac{1}{2} \nu_B - \nu_C = (1 - \alpha_D) \left(\frac{1}{2} \nu_B - \nu_C \right). \quad (48)$$

Substituting and using $m = \alpha_T(1 - \alpha_D)$:

$$d \log q_{AC} = \nu_C + m \left(\frac{1}{2} \nu_B - \nu_C - \frac{1}{2} \nu_B - \frac{1}{2} \nu_C - \frac{1}{2} d\tau \right) = \left(1 - \frac{3m}{2} \right) \nu_C - \frac{m}{2} d\tau. \quad (49)$$

As in (17), the tariff appreciates A in real terms directly through the price index (the $-\frac{m}{2}d\tau$ term), and the pass-through from the nominal rate is attenuated. Setting (49) to zero with $\nu_C = \frac{\rho - \rho^*}{3D} d\tau$ gives the real-rate reversal threshold: $(2 - 3m)(\rho - \rho^*) = 3mD$, and since $D = 3\alpha_D(\eta - 1) + \rho + 1$ this is linear in ρ , with solution

$$\boxed{\rho_q^* = \frac{(2 - 3m)\rho^* + 3m[3\alpha_D(\eta - 1) + 1]}{2 - 6m}} \quad (\rho_q^* > \rho^* \text{ for } m \in (0, \frac{1}{3})). \quad (50)$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $m = 0.08$): $\rho^* = 3.96$ and $\rho_q^* = 4.93$. Real depreciation against C requires more diversion than nominal depreciation, because part of the nominal movement is absorbed by the tariff's direct effect on A 's price index.

3 The N -country case

3.1 General linear structure

Countries $i \in \{1, \dots, N\}$, country 1 the numeraire, with the same technology and nested preferences as Section 2: home share α_{D_i} , origin weights β_{ij} , elasticities (η, ρ) . Let $\nu = (\nu_2, \dots, \nu_N)$ collect the log exchange rates and let a tariff $d\tau$ be imposed by country 1 on country 2. Linearizing the $N - 1$ independent trade-balance conditions at a free-trade baseline gives

$$J d\nu = -F d\tau \implies \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\text{adj}(-J) F}{\det(-J)}, \quad (51)$$

where $J_{ij} = \partial \text{TB}_i / \partial \nu_j$ and $F_i = \partial \text{TB}_i / \partial \tau$ at fixed exchange rates (the impact effects). The entries of J and F are linear in η and ρ , with baseline shares and incomes as coefficients, as in Section 2.4. At $N = 2$, (51) is the single equation (23).

Stability. The adjustment $\dot{\nu}_i \propto \text{TB}_i$ is locally stable if and only if all eigenvalues of J have negative real parts, which implies $\det(-J) > 0$: each response then has the sign of its numerator. The analogues at each scale:

$$N = 2 : \quad \varepsilon_X + \varepsilon_M > 1 \quad (21); \quad N = 3 : \quad D > 0; \quad \text{general } N : \quad \det(-J) > 0. \quad (52)$$

The multilateral Marshall–Lerner condition. Under gross substitutability, J has negative diagonal entries (the bilateral conditions of Section 1.6), nonnegative off-diagonal entries (another country’s appreciation improves your balance), and a dominant diagonal (homogeneity and Walras’ law). Then J is a stable Metzler matrix, $-J$ is an M-matrix, and

$$(-J)^{-1} \geq 0 \quad \text{elementwise.} \quad (53)$$

At the symmetric three-country point, $J = -\frac{mD}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: all of these conditions reduce to $D > 0$.

Result 3.1 (Generic sign rule). *At a stable equilibrium satisfying the multilateral Marshall–Lerner condition,*

$$\text{sign}\left(\frac{d\nu_i}{d\tau}\right) = \text{sign}\left(\sum_j w_{ij} F_j\right), \quad w_{ij} \equiv [(-J)^{-1}]_{ij} > 0. \quad (54)$$

In particular, if all impact effects share a sign, all responses share it. Sign ambiguity arises only when the impact vector is mixed.

The rule contains the earlier cases. At $N = 2$ there is one weight and one impact; the impact is positive by (22), so the currency appreciates unconditionally (Section 1). At $N = 3$,

$$(-J)^{-1} = \frac{4}{3mD} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \frac{d\nu_C}{d\tau} \propto F_B + 2F_C, \quad (55)$$

and ρ^* is the value of ρ at which $F_B + 2F_C = 0$.

F is linear in ρ and $\text{adj}(-J)$ has degree $N - 2$, so each numerator in (51) has degree $N - 1$ in ρ : a quadratic with one relevant root at $N = 3$ (the threshold of Section 2); degree three or more at $N \geq 4$, where single crossing can fail — configurations exist in which a bystander’s sign changes twice as ρ rises. The sign of any given response remains computable from (54).

3.2 The symmetric case: a closed form

Let all countries have equal size, common preferences, home share α_D , and equal origin weights $1/(N - 1)$. Country 1 (A) tariffs country 2 (B); the remaining $k = N - 2$ bystanders are identical, with representative c . By symmetry the system (51) reduces to two unknowns, (ν_B, ν_C) . Define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D), \quad D_N \equiv N\alpha_D(\eta - 1) + (N - 2)\rho + 1. \quad (56)$$

Solving the two-equation system (the derivation parallels Section 2.4 and is verified symbolically in the replication package) gives

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = - \frac{N\rho_1^* + (N-2)\rho}{N D_N} < 0, \quad (57)$$

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\rho - N\rho_1^*}{N D_N}, \quad (58)$$

$$\left. \frac{d \log e_{BC}}{d\tau} \right|_{\tau=0} = \frac{(N-1)\rho}{N D_N} > 0, \quad (59)$$

with $D_N > 0$ the stability condition; at $N = 3$, $D_3 = 3\alpha_D(\eta - 1) + \rho + 1 = D$. Hence:

Result 3.2 (*N-country threshold*). *In the symmetric N-country model, an isolated tariff depreciates A's currency against each untariffed bystander if and only if*

$$\boxed{\rho > \rho^*(N) = N[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]} \quad (60)$$

At $N = 3$ this is the threshold of Section 2: the coefficient 3 there is the number of countries. The threshold grows linearly in N because each bystander receives a share $1/(N-1)$ of the diverted expenditure.

For the symmetric trade war, add the responses to $d\tau_{AB}$ and its mirror $d\tau_{BA}$. Relabeling countries gives $\nu_C^{\text{war}} = 2\nu_C - \nu_B$, and substituting (57)–(58):

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0}^{\text{war}} = \frac{\rho - \rho_1^*}{D_N}, \quad \left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0}^{\text{war}} = 0. \quad (61)$$

Result 3.3 (*War threshold is independent of N*). *In a symmetric trade war, every bystander's currency appreciates against both belligerents if and only if*

$$\boxed{\rho > \rho_1^* = \frac{\rho^*(N)}{N}} \quad (62)$$

for any number of countries. With e_{AB} unchanged, each bystander's equation decouples and its sign equals the sign of its own impact effect, which does not dilute with N . At $\eta = 1$: $\rho_1^* = 1 - \alpha_T(1 - \alpha_D) < 1$, so any $\rho \geq 1$ gives joint depreciation; for $\eta > 1$, ρ_1^* can exceed one.

The formulas are verified against nonlinear equilibrium solves at $N = 4$ (threshold $\rho^*(4) = 5.28$ at $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, matching the numerical root to five decimal places).

3.3 Correspondence across N

The table collects each object and its analogue at every scale. $\rho_1^* = 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)$ throughout.

object	$N = 2$	$N = 3$ (symmetric)	general N
stability = ML	$\varepsilon_X + \varepsilon_M > 1$ (21)	$D > 0$	J stable; $\det(-J) > 0$
sufficient primitive	gross substitutes (ε 's ≥ 1)	$\eta \geq 1$	gross substitutes; $-J$ an M-matrix
sign rule	one impact, sign unconditional: ap- preciation	weights $\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ on two impacts	$\text{sign}(\sum_j w_{ij} F_j)$, $w > 0$
isolated-tariff threshold	none (no bystander exists)	$\rho^* = 3\rho_1^*$	$N\rho_1^*$ (symmetric); roots of a degree- $(N-1)$ poly- nomial in general
war threshold	—	$\rho_1^* = \rho^*/3$	ρ_1^* , independent of N

4 Canonical PTA model: competing exporters

4.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y . Endowments: A holds only y ; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \quad u'_i > 0, \quad u''_i < 0. \quad (63)$$

A levies specific tariffs t_B, t_C on imports of x from B and C . Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A . Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \quad (64)$$

4.2 Demands and market clearing

From $u'_i(x_i)$ = (local price of x):

$$M_A(p) = x_A^d(p), \quad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \quad (65)$$

with $M'_A < 0$ and $E'_j > 0$. Market clearing for x :

$$M_A(p) = E_B(q_B) + E_C(q_C) = E_B(p - t_B) + E_C(p - t_C). \quad (66)$$

Equation (66) determines p given (t_B, t_C) ; then q_B, q_C follow from (64).

4.3 Comparative statics of a discriminatory tariff

Totally differentiate (66) with respect to t_B , holding t_C :

$$M'_A \frac{dp}{dt_B} = E'_B \left(\frac{dp}{dt_B} - 1 \right) + E'_C \frac{dp}{dt_B} \implies \frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1). \quad (67)$$

Therefore

$$\frac{dq_B}{dt_B} = \frac{dp}{dt_B} - 1 = \frac{E'_C - M'_A}{E'_B + E'_C - M'_A} \cdot (-1) < 0, \quad (68)$$

$$\frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (69)$$

Result 4.1 (Viner–Mundell). *A discriminatory tariff on B worsens B’s terms of trade and improves C’s terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B’s and worsens C’s terms of trade.*

4.4 The PTA assumptions as restrictions on the three-country threshold

An improvement in C ’s terms of trade is a rise in e_{AC} (Section 2.3). The canonical model signs this unconditionally from $E'_j > 0$ and $M'_A < 0$; Section 2 requires $\rho > \rho^*$. The difference is the three pieces of the threshold (benchmark values: $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$):

$$\rho^* = \underbrace{3}_{B-C \text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness credit}} = 3.0 + 1.2 - 0.24 = 3.96. \quad (70)$$

The linkage claim collects the two forces running through B – C trade: B ’s income loss cuts its purchases from C , and C substitutes toward B ’s cheapened goods. Each canonical assumption removes one piece. Recomputed thresholds under each restriction (severing trade changes the baseline, so the pieces are approximately additive):

restriction	threshold ρ^*	piece removed
none (benchmark)	3.96	—
no B – C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage claim
no home tradables ($\alpha_D = 0$)	1.80	the home-switching claim
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized (small alone)
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B – C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

Three readings:

- The largest piece is the B – C linkage: severing it alone moves the threshold from 3.96 to 1.05.
- Non-tradables are not an independent force: with the linkage severed and $\alpha_D = 0$, no threshold exists at any α_T . The non-tradable share enters only through the credit.
- Two routes reach $\rho^* \leq 0$: $\{\alpha_D = 0, \alpha_T = 1\}$ or $\{\alpha_D = 0, \beta_{BC} = \beta_{CB} = 0\}$. The canonical model takes both, adds quasilinearity, and in the Viner case sets $\rho = \infty$.

Result 4.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho^* \leq 0$. The unconditional Viner–Mundell sign is the threshold condition of Section 2 evaluated where it cannot bind.*

The literature maps onto the table:

- competing exporters (Bagwell–Staiger 1999; Ornelas 2005): $\beta_{BC} = \beta_{CB} = 0$, quasilinear;
- symmetric endowment blocs (Krugman 1991; Bond–Syropoulos 1996): $\alpha_D \approx 0$, $\alpha_T = 1$, one elasticity;
- Viner: $\rho = \infty$;
- Kemp–Wan 1976: none of these restrictions, and no sign claims.

Applied models (GTAP-class CGE, with a two-level Armington nest as standard practice; quantitative trade) keep all pieces of (70) positive and produce configuration-dependent signs, but report simulations, not the condition. The threshold of Section 2 is that condition.